Non-Linear and Additive Models

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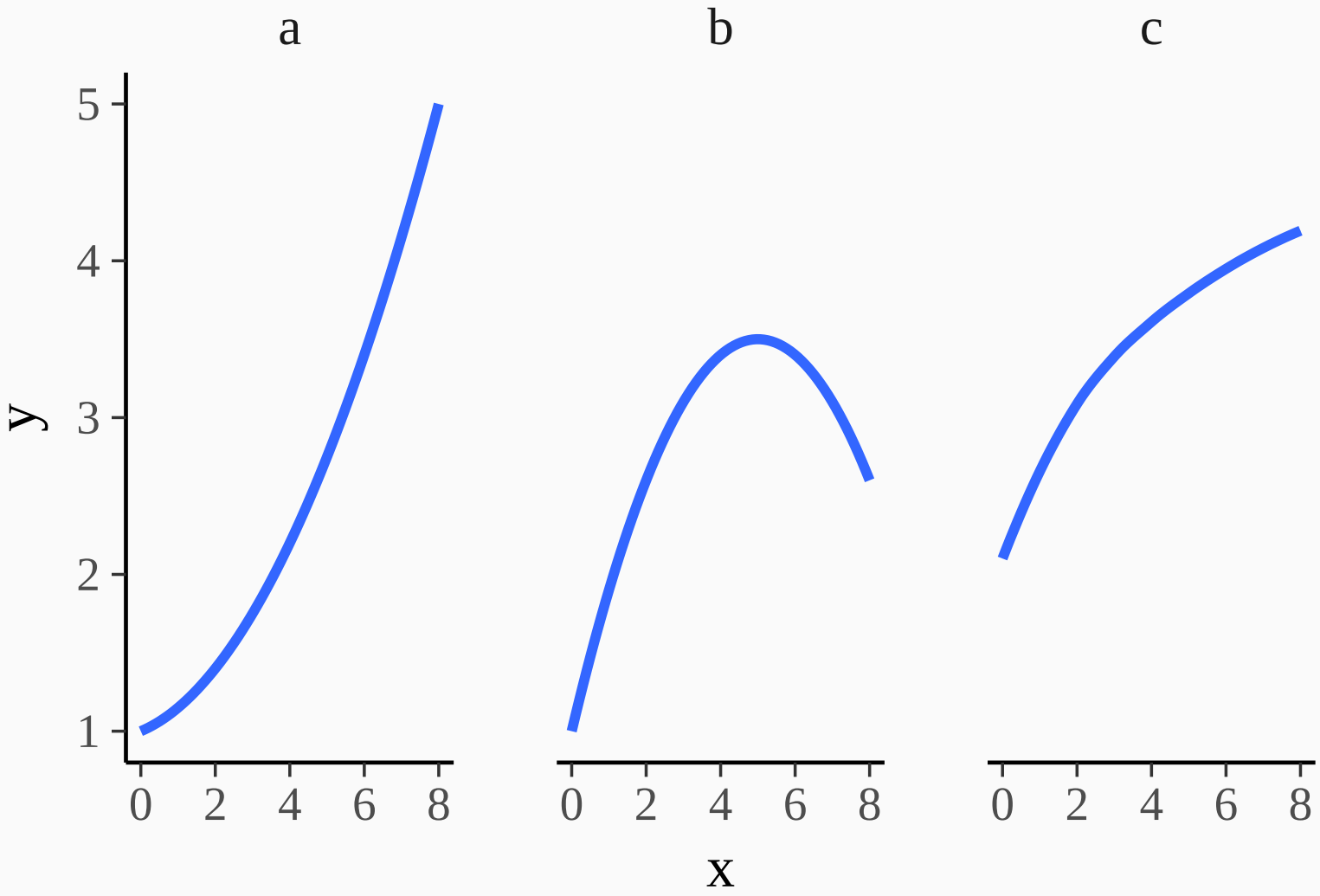
Non-linear growth

Growth over time is not always linear



Three Approaches to Non-Linearity

- Transform variables so that the relationship is linear
- Explicitely specify a parametric non-linear model
- Estimate the non-linear function from the data using semi-parametric approaches such as splines or Gaussian processes

Approach 1: Transform Predictor Variables

Many non-linear relationships can be modeled with linear regression

Quadratic Regression:

$$y = b_0 + b_1 x + b_2 x^2$$

Cubic Regression:

$$y = b_0 + b_1 x + b_2 x^2 + b_3 x^3$$

Approach 1: Transform Response Variables

Suppose that the relationship between y and x is

$$y = \exp(b_0 + b_1 x)$$

Then the relationship is linear on the log scale of y:

$$\log(y) = \log(\exp(b_0 + b_1 x)) = b_0 + b_1 x$$

Common response Variable transformations

- **log** for right skewed and strictly positive responses
- **sqrt** or **log1p** for right skewed and non-negative responses
- **logit** for probabilities
- **Fisher's Z** for correlations
- **Box-Cox** tranformation to stabilize variances

Time for exercise

'variable_transformations.R'

Approach 2: Explictely Specify a Non-Linear Model

Some relationships cannot be rewritten as a linear predictor:

$$y = a \exp(-(bx)^c)$$

where a, b, and c are non-linear parameters.

Solution: Define linear predictors for the non-linear parameters

$$a = \eta_a, \quad b = \eta_b, \quad c = \eta_c$$

Example: Cumulative Insurance Loss Payments

- Insurance companies initially earn money from insurances
- Over the years, part of the money has to payed to the insured
- Question: How to predict the cumulative loss payments originated from different origin years?

Cumulative Insurance Loss Payments: Data

```r
loss <- read.csv("data/loss.csv")
head(loss, 10)
```

X	AY	dev	cum
1	1991	6	357.848
2	1991	18	1124.788
3	1991	30	1735.330
4	1991	42	2182.708
5	1991	54	2745.596
6	1991	66	3319.994
7	1991	78	3466.336
8	1991	90	3606.286
9	1991	102	3833.515
10	1991	114	3901.463

Cumulative Insurance Loss Payments: The Model

Use a Weibull distribution function to model the growth in loss payments:

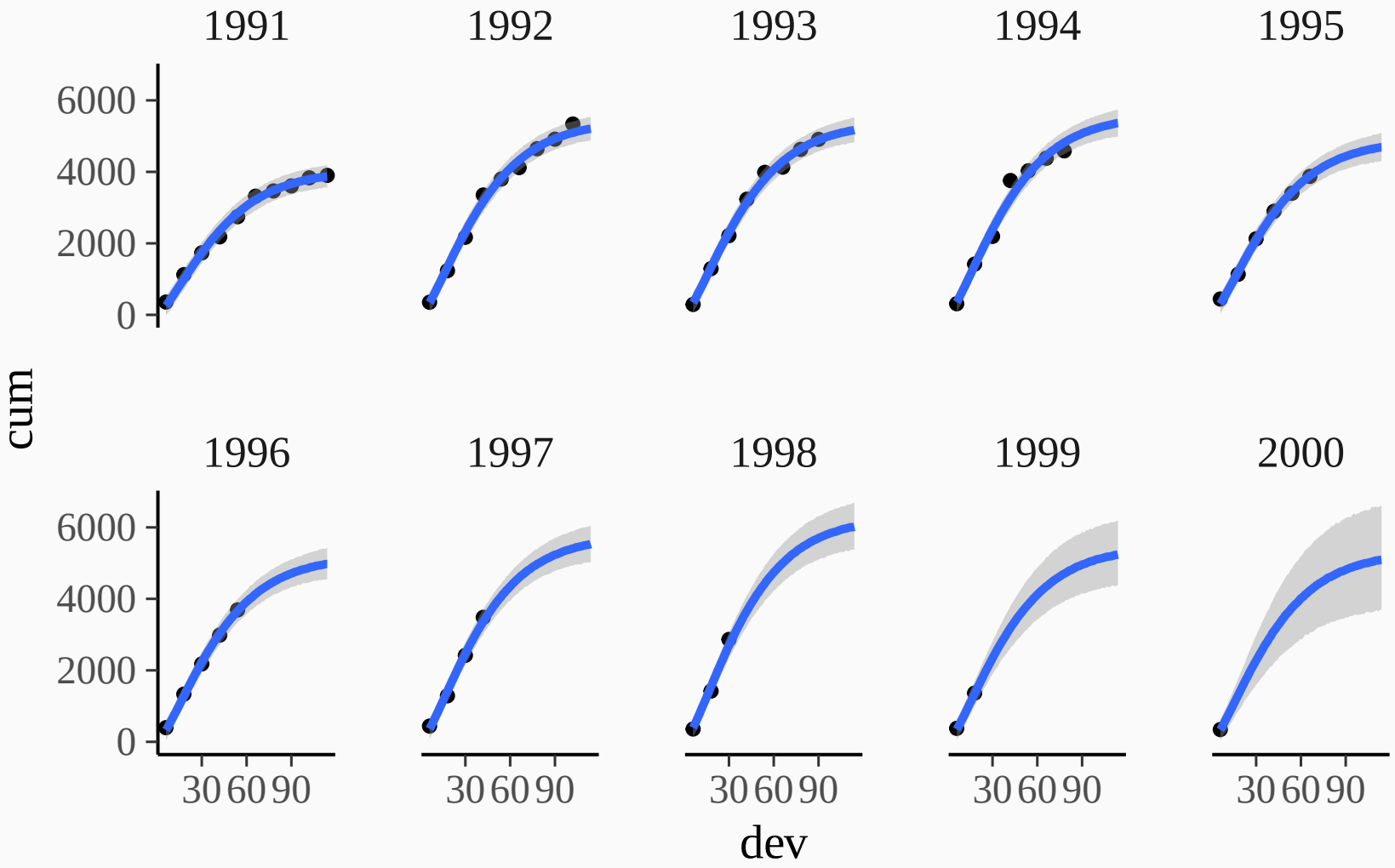
$$cum = ult \times f(dev|\omega, \theta) = ult \times \left(1 - \exp\left(-\left(\frac{dev}{\theta}\right)^{\omega}\right)\right)$$

- Variable *dev* is the deviation between orginal year and the year at which loss payments are evaluated.
- The non-linear parmater are the ultimate loss *ult*, as well as ω and θ
- The estimated ultimate loss is what we are primarily interested in

Cumulative Insurance Loss Payments: Model Fitting

```r
fit_loss <- brm(
  bf(cum ~ ult * (1 - exp(-(dev/theta)^omega)),
     ult ~ 1 + (1|AY), omega + theta ~ 1, nl = TRUE),
  data = loss, family = gaussian(),
  prior = c(set_prior("normal(5000, 1000)", nlpar = "ult"),
            set_prior("normal(1, 2)", nlpar = "omega"),
            set_prior("normal(45, 10)", nlpar = "theta"))
)
```

Cumulative Insurance Loss Payments: Visualization



Cumulative Insurance Loss Payments: Expected Ultimate Loss

```r
round(coef(fit_loss)$AY[, , "ult_Intercept"])
```

```
##      Estimate Est.Error 2.5%ile 97.5%ile
## 1991     4023       112    3812     4262
## 1992     5409       154    5136     5732
## 1993     5367       167    5062     5722
## 1994     5569       190    5221     5978
## 1995     4868       190    4512     5266
## 1996     5171       218    4773     5633
## 1997     5742       266    5246     6290
## 1998     6255       338    5614     6935
## 1999     5445       468    4576     6401
## 2000     5298       757    3852     6887
```

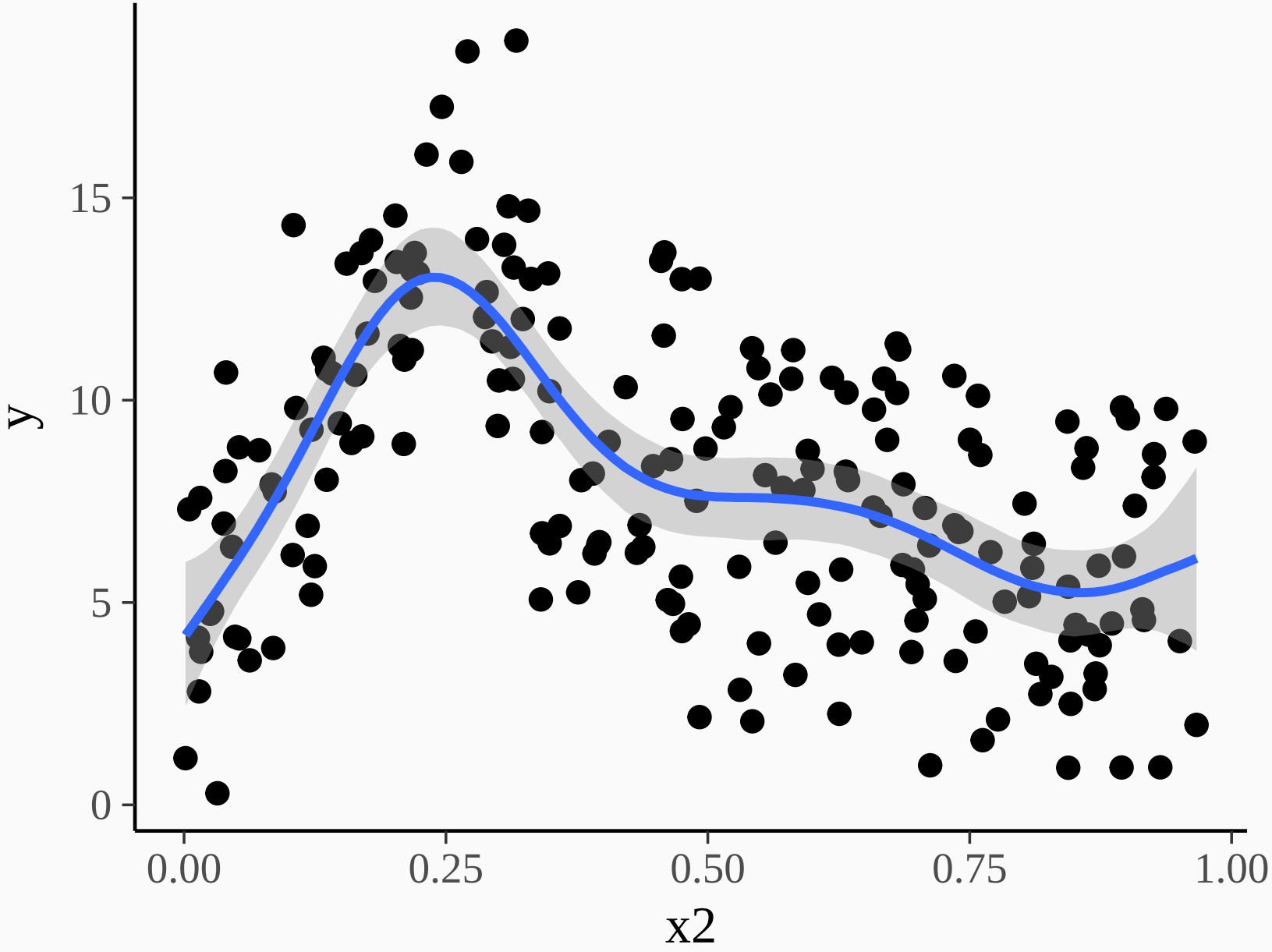
Approach 2: Potential Problems

- Specification of an appropriate non-linear function
- Invalid numerical computations
- Non-identified models
- Convergence problems

Time for exercise

'non_linear_models.R'

Approach 3: Smooth Terms (Additive Models)



Splines: Introduction

$$y = b_0 + f(x)$$

Goal: Estimate the non-linear function f without having any idea about its form a-priori

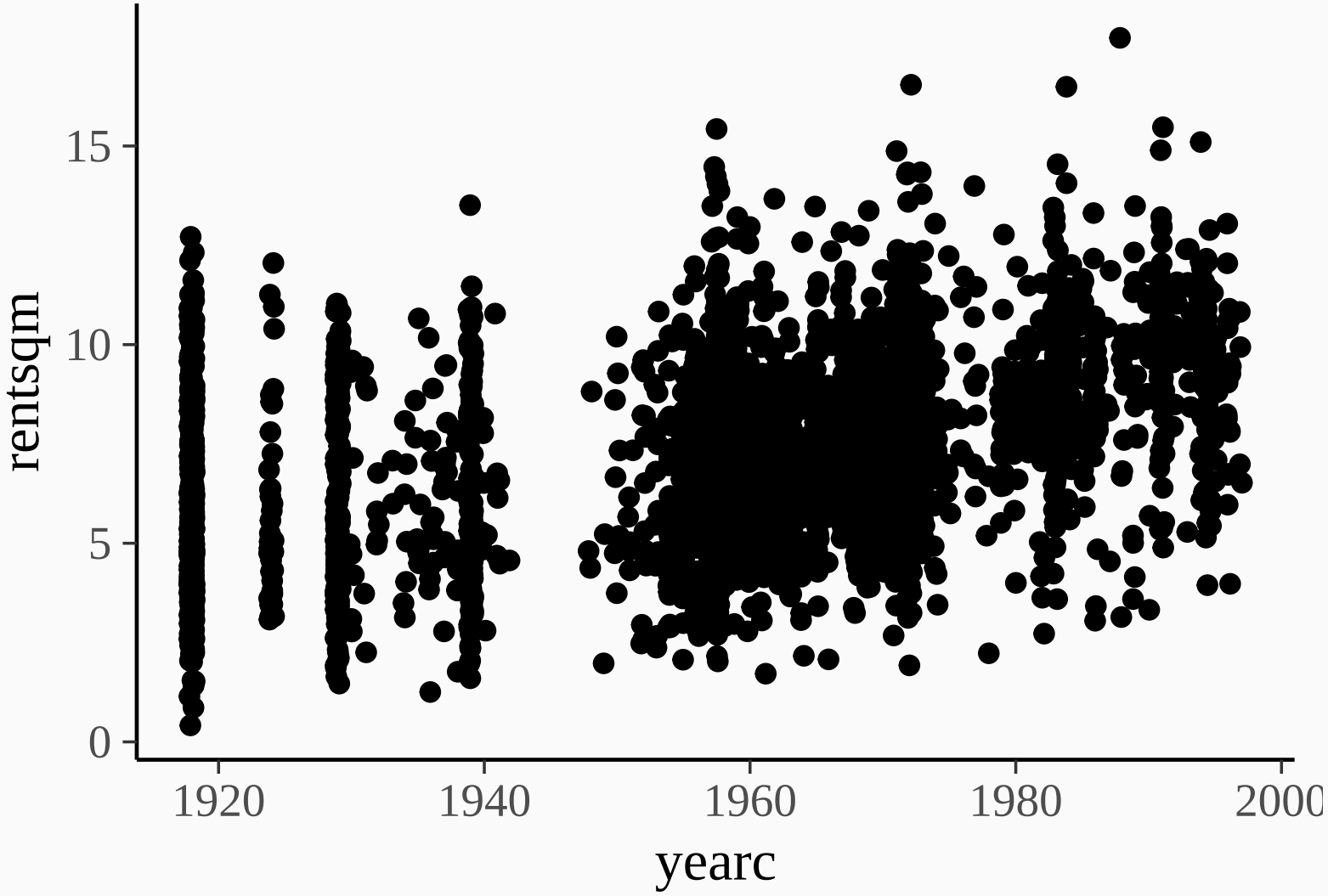
Solution: Define a set of functions smoothly connected at their intersections (i.e. use splines)

Example: Housing Rents in Munich

Data described in Fahrmeier et al. (2013)

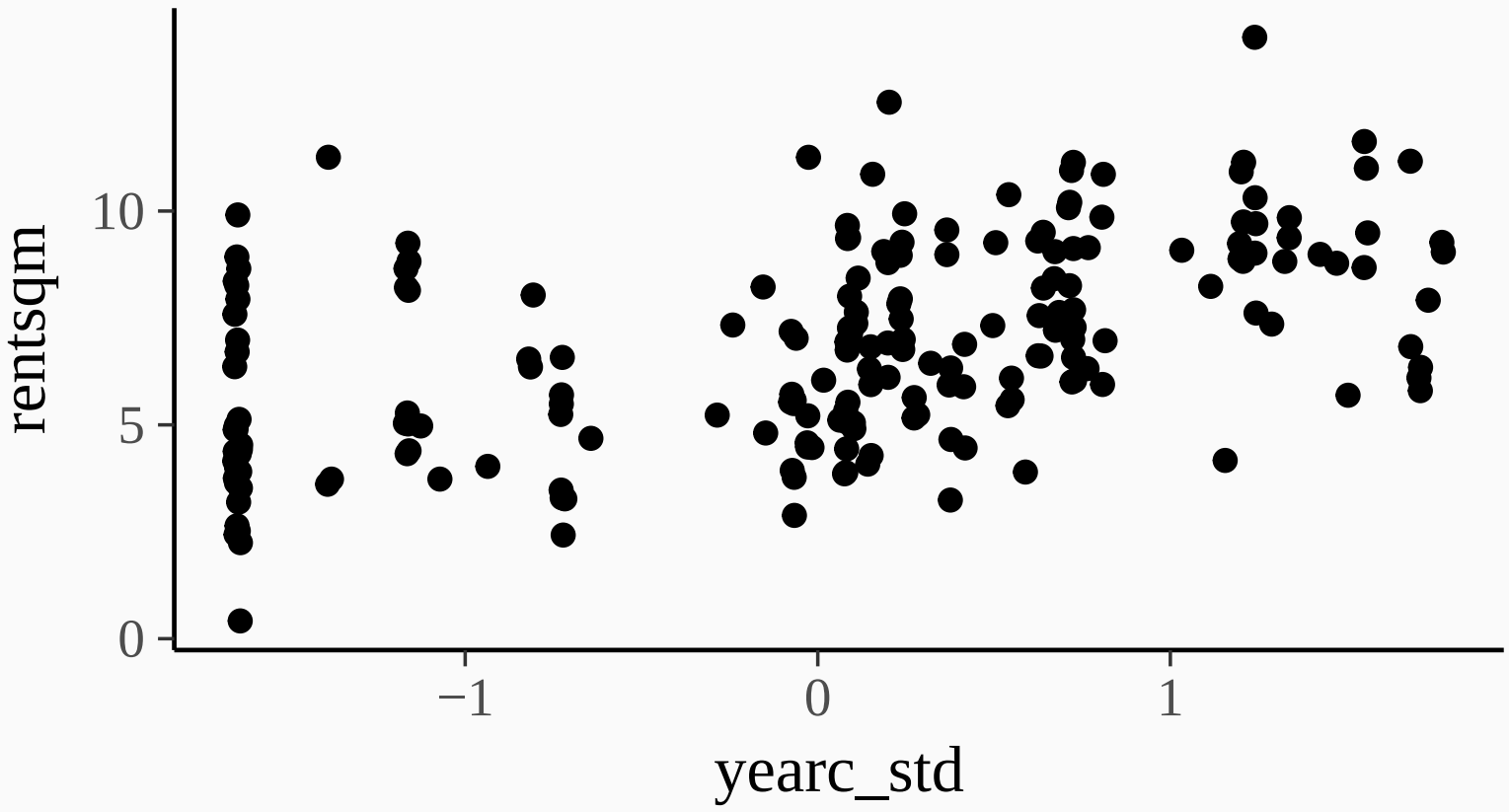
rent	rentsqm	area	yearc
109.9487	4.228797	26	1918
243.2820	8.688646	28	1918
261.6410	8.721369	30	1918
106.4103	3.547009	30	1918
133.3846	4.446154	30	1918
339.0256	11.300851	30	1918
215.2308	6.942928	31	1918
323.2308	10.426800	31	1918
216.3077	6.759615	32	1918
245.2820	7.432790	33	1918

Housing Rents: Visualization



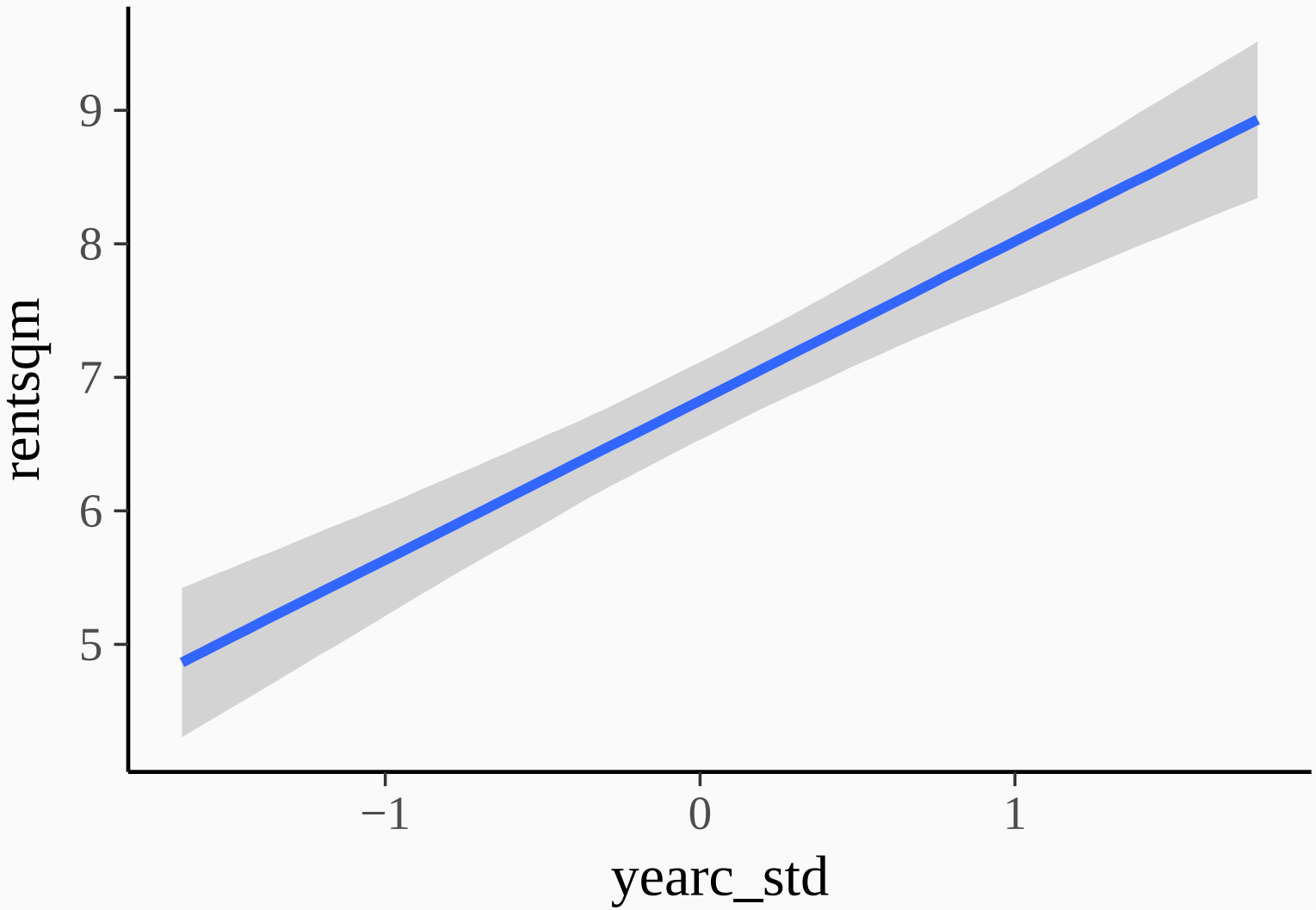
Housing Rents: Subset

Take a random subset for better illustration:



Housing Rents: Linear model

```
fit_rent1 <- brm(rentsqm ~ yearc_std, data = srent99)
```



Defining Linear Splines

Define a *knot* k where the linear slope changes

We choose $k = -0.2$:

$$\text{rentsqm} = b_0 + b_1 \text{year} + b_2(\text{year} - (-0.2))_+$$

where

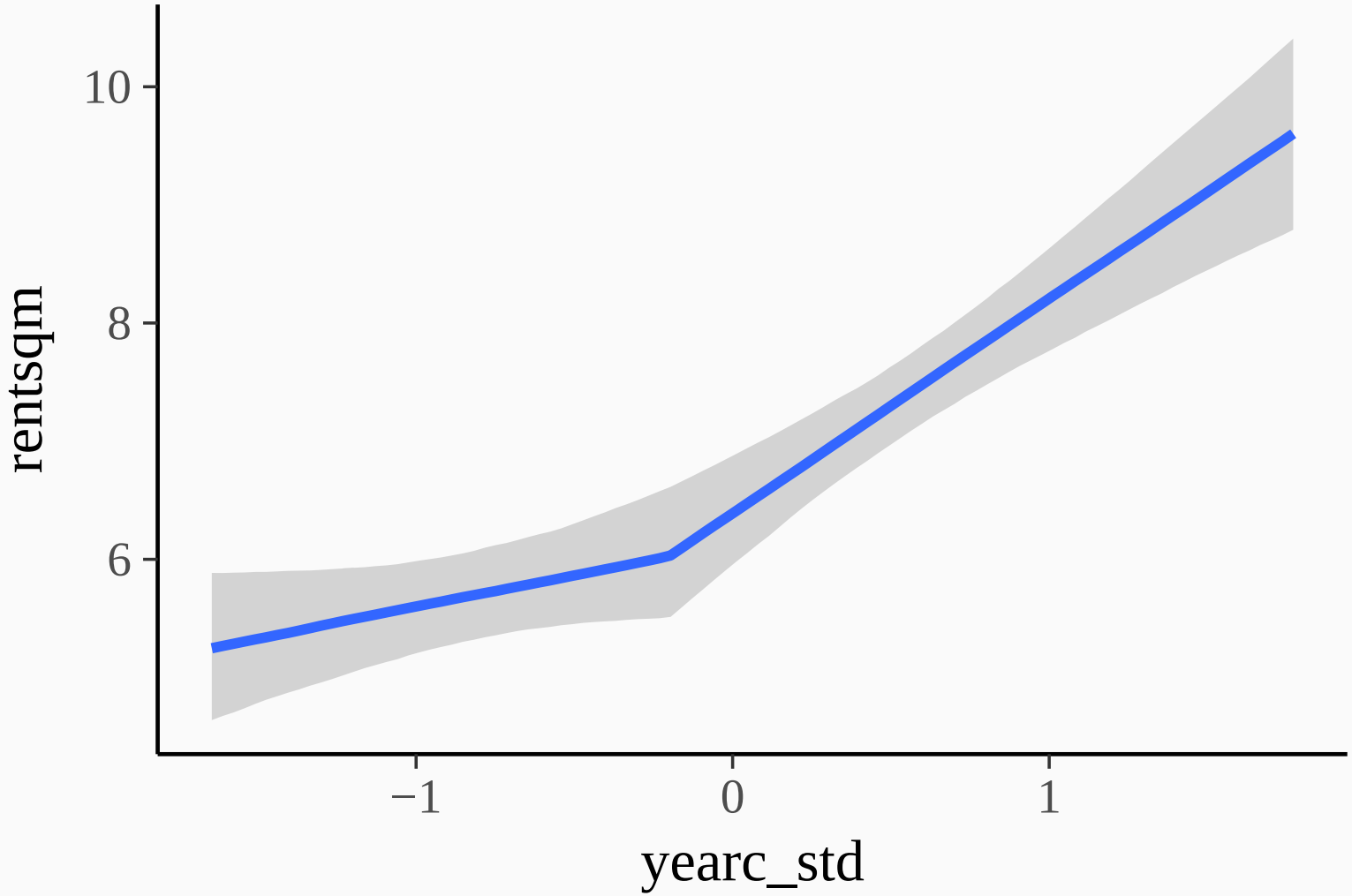
$$(x)_+ = 0 \text{ if } x < 0$$
$$(x)_+ = x \text{ if } x \geq 0$$

As a result, we have defined

$$f(\text{year}) = b_1 \text{year} + b_2(\text{year} - (-0.2))_+$$

Housing Rents: Linear Splines

```
rhs <- function(x, TH) ifelse(x >= TH, x - TH, 0)
fit_rent2 <- brm(rentsqm ~ yearc_std + rhs(yearc_std, -0.2), data = srent99)
```



Defining Splines with More Knots

Add multiple terms of the form $(\text{year} - k_i)_+$:

$$\text{rentsqm} = b_0 + b_1\text{year} + b_2(\text{year} - k_1)_+ + b_3(\text{year} - k_2)_+ + b_4(\text{year} - k_3)_+$$

More generally, a linear spline can be written as

$$f(x) = \sum_{i=1}^{K} b_i(x - k_i)_+$$

Define Cubic Splines

Linear Splines are not smooth at the internal knots

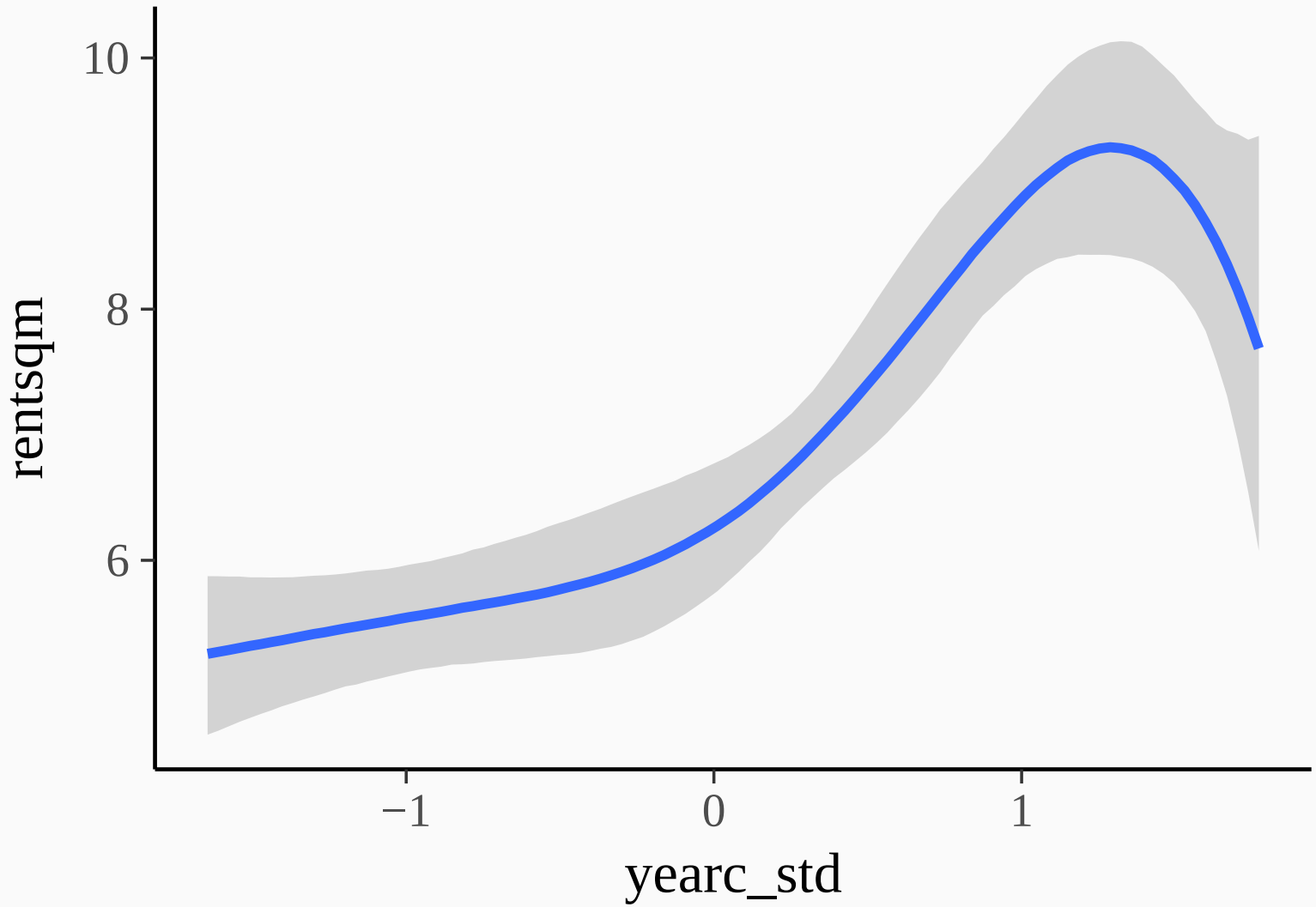
To improve smoothness, we have to use higher order terms for instance cubic splines:

$$f(x) = \sum_{i=1}^{K} b_i (x - k_i)_+^3$$

Cubic Splines ensure that the first and second derivatives of the function match at the knots.

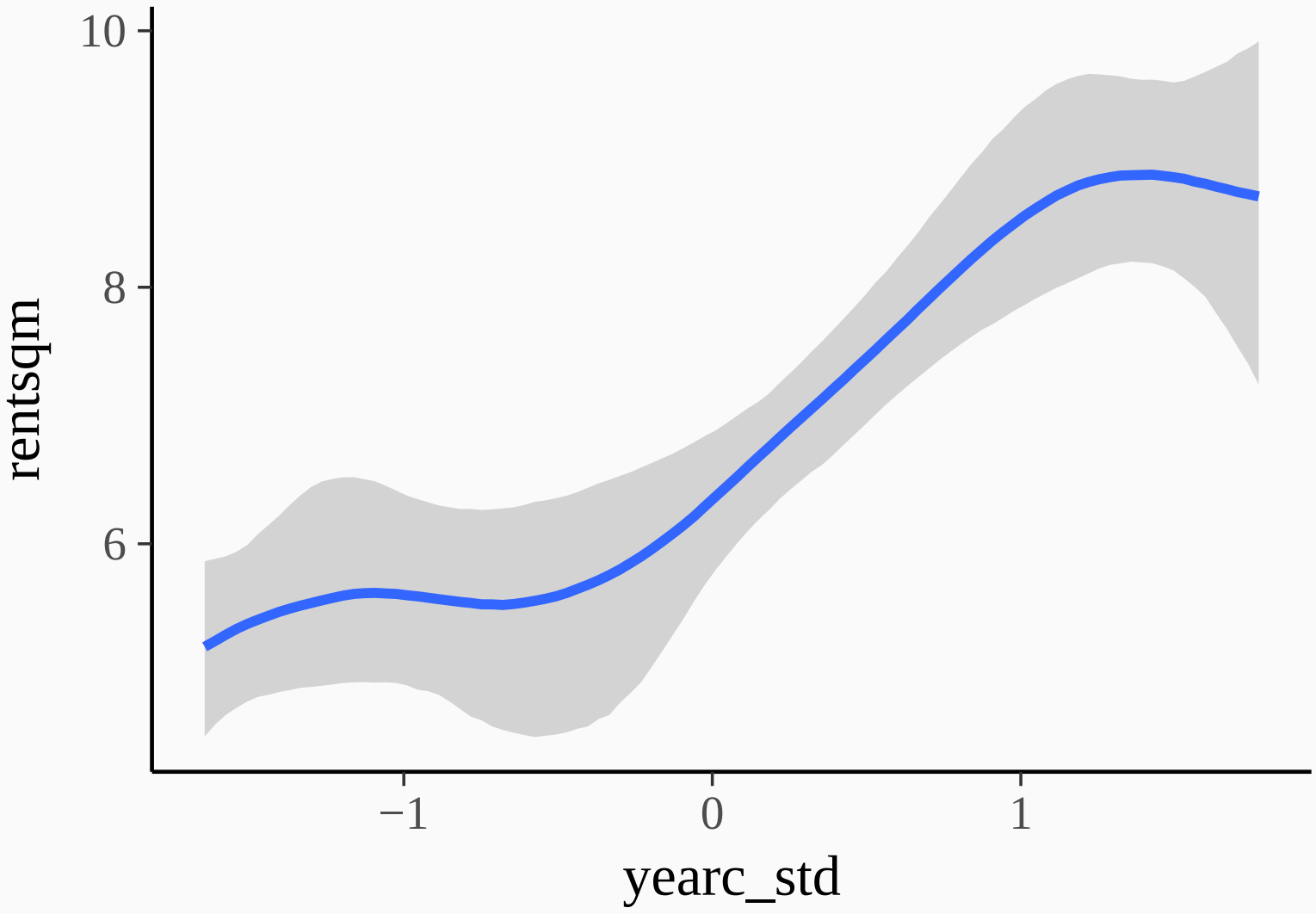
Housing Rents: Cubic Splines

```
fit_rent4 <- brm(rentsqm ~ yearc_std + I(rhs(yearc_std, -0.75)^3) +
                    I(rhs(yearc_std, 0.13)^3) + I(rhs(yearc_std, 0.72)^3),
                 data = srent99)
```



Housing Rents: Setting Up Splines Automatically

```
fit_rent5 <- brm(rentsqm ~ s(yearc_std), data = srent99)
```



Approach 3: Potential Problems

- Uninterpretable model parameters
- Flexibility of splines may lead to overfitting
- Prediction beyond the original data space is difficult

Further Reading

- Zuur A. F. (2012). A beginner's guide to generalized additive models with R. Newburgh: Highland Statistics.
- Zuur, A. F., Saveliev, A. A., & Ieno E. N. (2014). A beginner's guide to generalized additive mixed models with R. Newburgh: Highland Statistics.

Time for exercise

'additive_models.R'